

A
MINI PROJECT REPORT
ON

SMART TROLLEY WITH AUTOMATIC
BILLING SYSTEM

SUBMITTED TO THE DEPARTMENT OF
ELECTRONICS & COMMUNICATION ENGINEERING

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Session: 2024-25

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ABSTRACT

The Smart Trolley with an Automatic Billing System is an innovative solution designed to enhance the shopping experience by automating product scanning, billing, and payment. Using technologies like RFID, barcode scanning, weight sensors, and IoT integration, the system tracks items in real-time, updating the total bill as products are added. This eliminates the need for manual scanning and speeds up the checkout process, providing a more efficient and accurate shopping experience.

The system includes a touchscreen display to show item details, prices, and total costs, while integrated digital payment methods allow customers to pay directly through the trolley, supporting contactless and mobile payments. Additionally, IoT connectivity allows real-time inventory management, enabling retailers to track stock levels more accurately and improve operational efficiency.

Further enhancing the user experience, technologies like computer vision and optional voice recognition enable a more intuitive and hands-free interaction with the system. Overall, the Smart Trolley with Automatic Billing System streamlines the shopping process, reduces human errors, and offers a more convenient and efficient experience for both customers and retailers.

CHAPTER ONE

SMART TROLLEY WITH AUTOMATIC BILLING SYSTEM – AN OVERVIEW

1.1 INTRODUCTION

Abstract The Smart Trolley with Automatic Billing System is an innovative system that aims to revolutionize the overall shopping process by making it fast, smooth and efficient. Using advanced technologies such as RFID, IoT, and barcode scanner, the smart trolley allows customers to scan items that they add, instantly updating their total bill and displaying it on an interactive screen. The trouble of standing in long queues while checking out, charging incorrect amounts, and so forth is completely eliminated with this arrangement and makes the shopping process quicker and smoother. More than just convenient for consumers, a mobile app brings many benefits for retailers with better inventory management, contactless payment handling and improving customer satisfaction. This smart trolley is a move towards the future of the retail sector, offering an integrated shopping experience with technology and convenience. This report discusses what its purpose is, why do we need this, its features, advantages, uses, and the evolved techniques that make it a game-changer in the industry.

1.2 WHAT IS AUTOMATION?

Automation is technology that performs tasks or processes without human assistance. It refers to the design and use of any system, machine or software that can work independently of an operator, based on pre-designed software or algorithms. Automation aims to boost efficiency, minimize human error, and execute repetitive or intricate activities more rapidly and consistently than a human can.

Automation is also known and used in fields like manufacturing, data processing, service provider, and even in daily type tools like smart home devices. This saves time & reduces costs & boosts productivity by automating tasks, such as production line processes, billing & even data analysis. In some simple terms of everyday life; Automation is seen in self-checkout kiosks, robotic vacuum cleaners, smart home devices which adjust lighting, temperature, etc.

1.3 WHAT TYPE OF AUTOMATION USED IN SMART TROLLEY?

The smart trolley incorporates several types of automation to enhance the shopping experience:

I. RFID (Radio Frequency Identification) Automation

RFID technology is used to automatically detect and track items as they are added to the trolley. RFID tags attached to products are scanned by antennas in the trolley, updating the bill in real-time without requiring manual scanning.

II. Barcode Scanning Automation

In addition to RFID, smart trolleys often use barcode scanning automation. As customers place items in the trolley, built-in barcode scanners automatically scan the product's barcode, updating the total cost on the display screen.

III. Weight Sensors Automation

Weight sensors in the trolley are used to verify the items being added, ensuring the correct items are tracked and billed. These sensors can also help detect if an item is removed or replaced.

IV. Payment Automation

Payment automation is integrated into smart trolleys to enable seamless transactions. Once the customer is ready to checkout, they can make a payment through digital or contactless payment systems directly through the trolley.

V. Inventory Management Automation

The trolley can be connected to the store's inventory system through IoT (Internet of Things), automatically updating inventory as items are added or removed. This helps retailers manage stock levels in real-time.

VI. User Interface Automation

Smart trolleys are equipped with an interactive display or touch screen that automates the user interface for customers. It displays item details, prices, discounts, and the total bill, allowing for a smooth, self-guided shopping experience.

These automation technologies work together to make the shopping process faster, more accurate, and more efficient for both customers and retailers.

1.4 BENEFITS OF SMART TROLLEY WITH AUTOMATIC BILLING SYSTEM

Time efficiency

The smart trolley significantly reduces the time spent at checkout by allowing customers to scan items as they shop. The real-time updating of the bill ensures a faster, seamless experience, eliminating long queues at traditional checkout counters.

Accuracy and Reduced Errors

With automatic scanning and real-time updates, the risk of human error in pricing and billing is minimized. This ensures that customers are charged correctly for the items they purchase, improving trust and satisfaction.

Enhanced customer experience

The convenience of instant billing and payment options enhances the overall shopping experience. Customers can focus more on their shopping without worrying about waiting in line or facing delays, resulting in a smoother, more enjoyable store visit.

Contactless Payment Option

The smart trolley supports various payment methods, including contactless payment options, enhancing security and reducing the need for physical interactions during the checkout process.

Reduced Queue Congestion

With real-time billing and automated payments, customers can bypass traditional checkout lines, reducing congestion at cash registers and improving the flow of customers within the store.

1.5 PURPOSE OF SMART TROLLEY WITH AUTOMATIC BILLING SYSTEM

The primary purpose of the Smart Trolley with Automatic Billing System is to enhance the shopping experience by automating the traditionally manual billing process, making it faster, more accurate, and more convenient. By integrating advanced technologies such as RFID, IoT, and barcode scanning, this system allows customers to scan their items in real time as they shop and instantly see the total bill on a display screen.

This system aims to:

- I. Reduce Checkout Time:** By eliminating the need for traditional checkout lines, the smart trolley streamlines the shopping process, allowing customers to complete their purchases quickly and easily.
- II. Improve Billing Accuracy:** Automation reduces human errors during billing, ensuring customers are charged correctly for their items.
- III. Enhance Customer Convenience:** The system provides an effortless shopping experience, offering real-time updates and payment options that minimize wait times and enhance satisfaction.
- IV. Increase Operational Efficiency for Retailers:** Retailers can manage inventory in real-time and reduce the reliance on cashier staff, optimizing store operations and focusing resources on improving customer service.

Ultimately, the purpose of the smart trolley is to transform the retail industry by providing a modern, efficient, and user-friendly solution that benefits both customers and retailers.

1.6 APPLICATIONS OF SMART TROLLEY WITH AUTOMATIC BILLING SYSTEM

Supermarkets and Grocery Stores

Smart trolleys can be used in supermarkets and grocery stores to streamline the shopping and checkout process, allowing customers to scan items as they shop and complete their purchases quickly.

Retail Stores and Malls

In retail stores and malls, smart trolleys enhance the shopping experience by enabling real-time item scanning, automatic billing, and seamless payment processing.

Airports and Duty-Free Shops

Smart trolleys in airports and duty-free shops allow travellers to shop efficiently, bypass long checkout lines, and complete transactions in a timely manner.

Wholesale and Hypermarkets

Wholesale stores and hypermarkets can benefit from smart trolleys to handle large volumes of products, improve inventory management, and speed up the billing process.

Self-Checkout Solutions

Smart trolleys can serve as an alternative to traditional self-checkout kiosks, offering customers a mobile, interactive solution for scanning and payment.

These applications highlight how smart trolleys can improve efficiency and customer satisfaction across a range of retail and service environments.

1.7 DIFFERENT TECHNIQUES OF SMART TROLLEY WITH AUTOMATIC BILLING SYSTEM

RFID Technology

RFID (Radio Frequency Identification) is one of the core technologies used in smart trolleys. Each item in the store is equipped with an RFID tag, which contains product information. As customers add products to their trolley, RFID readers built into the trolley automatically detect and record the items. The system updates the total bill in real-time without the need for

manual scanning, making the shopping experience more seamless and efficient. This technique eliminates the need for customers to scan individual items at checkout, speeding up the entire process.

Barcode Scanning

Barcode scanning is another widely used technique in smart trolleys. Each product in the store has a unique barcode, and the trolley is equipped with built-in barcode scanners that automatically read these barcodes as items are placed inside. This system ensures that all items are accurately detected and added to the bill. It's a cost-effective solution for smart trolleys and is particularly useful in stores where products may not have RFID tags. Barcode scanning offers a reliable way to maintain billing accuracy while reducing human error.

Weight Sensors

Weight sensors are integrated into the trolley to track the weight of items placed inside. These sensors help confirm that the correct items are added to the trolley by comparing their weight against the expected weight based on the product type. Weight sensors also assist in detecting if an item is removed or replaced, ensuring that the billing system is always updated accurately. This technique adds an extra layer of validation to the scanning process, ensuring that the items in the trolley correspond to the total bill.

Touchscreen Interface

Many smart trolleys are equipped with a touchscreen interface that displays key information to the customer. The interface shows the details of each item added to the trolley, along with the price and any discounts applied. It also displays the total bill in real-time, giving customers the ability to track their purchases as they shop. The touchscreen can be interactive, allowing customers to make changes, view offers, or even check the availability of items in the store. This intuitive user interface enhances the overall shopping experience by providing customers with control over their purchases.

IoT Integration

The Internet of Things (IoT) plays a critical role in the smart trolley's operation. IoT technology enables the trolley to communicate with the store's central system in real time, updating inventory and billing information. As items are added to the trolley, the IoT system ensures that stock levels are accurately updated in the store's database, preventing discrepancies between the inventory on the shelves and what is being purchased. This integration also allows for better management of product availability, promotions, and customer data, improving the overall retail experience for both customers and store operators.

Payment Integration

One of the most significant features of a smart trolley is its ability to integrate payment systems. Once the shopping is complete, the customer can pay directly through the trolley, which can support various payment methods, including contactless payments, mobile wallets, and credit card processing. The system automatically processes the payment, updating the transaction in real-time. This eliminates the need for customers to wait in line at checkout counters, enhancing convenience and reducing friction in the shopping experience.

Voice Recognition

Voice recognition technology is used to make the smart trolley even more user-friendly. Customers can interact with the trolley using voice commands, asking for information about their items, total bill, or current promotions. The trolley can provide responses through speakers, allowing customers to get real-time feedback without needing to manually interact with the touchscreen or system. This hands-free interaction enhances accessibility, especially for customers with disabilities or those who prefer voice-based commands. It also makes the shopping experience more intuitive and effortless.

These techniques, when combined, create a highly automated and efficient system that improves the shopping process. From automatic item detection and real-time billing to seamless payment processing, these technologies work together to offer a smoother, faster, and more convenient shopping experience for customers while enhancing operational efficiency for retailers.

1.8 NEED OF SMART TROLLEY WITH AUTOMATIC BILLING SYSTEM

The need for the smart trolley with an automatic billing system is driven by the demand for faster, more efficient shopping experiences. It reduces checkout times by allowing customers to scan and update their bills in real time, eliminating long waits at traditional counters. The system also minimizes human errors in billing, ensuring accurate transactions. With labor shortages common in retail, smart trolleys reduce reliance on cashiers, freeing up staff for other tasks. Additionally, they cater to the growing preference for digital and contactless payments while providing real-time inventory updates, enhancing both customer satisfaction and operational efficiency.

In this chapter, we discussed the applications, needs, purpose, and techniques of the Smart Trolley with an Automatic Billing System. The next chapter will focus on the components used in the system, exploring the hardware and software involved. It will also include a block diagram to illustrate the system's working.

CHAPTER TWO

TECHNOLOGY USED

2.1 INTRODUCTION

The smart trolley with an automatic billing system incorporates various advanced technologies to enhance the shopping experience for customers and streamline operations for retailers. These technologies work together to automate the scanning, tracking, and billing of products, eliminating the need for manual processes and reducing errors. By integrating systems such as RFID, barcode scanning, weight sensors, IoT, and payment solutions, smart trolleys provide an efficient and seamless shopping experience. Additionally, technologies like computer vision and voice recognition further improve the ease of use and interaction, making the entire process faster, more accurate, and more convenient. These technologies not only meet the demands of modern consumers but also help retailers optimize their inventory management and enhance overall operational efficiency.

2.2 BLOCK DIAGRAM

The block diagram for the circuit is shown in Fig. 2.1.

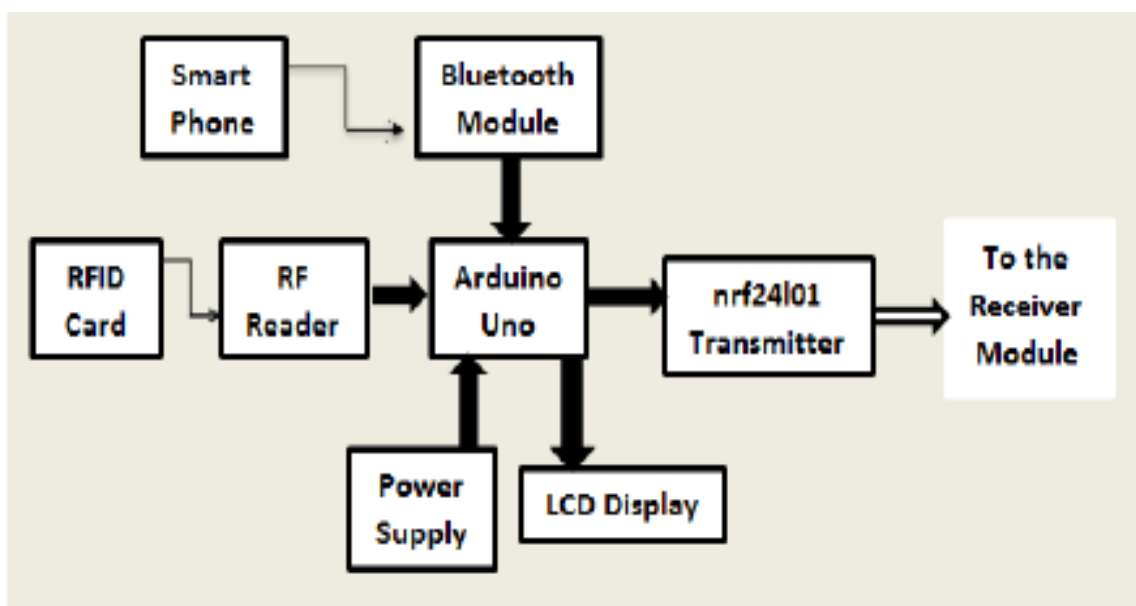


Fig. 2.1 Block Diagram

2.3 COMPONENT LIST

Here we explain the components used to design the whole system with detailed specifications. Technology used and particulars are also given regarding each of the components. The hardware developed for the present system consists of the following main modules and circuitry:

- Arduino Nano Microcontroller board
- SIM 800L
- LCD
- Buzzer
- Potentiometer
- Buck Convertor
- EM18 Module Reader
- LEDs

2.4 ARDUINO NANO MICROCONTROLLER BOARD

The **Arduino Nano** is a compact microcontroller board based on the **ATmega328P**, ideal for small and efficient electronics projects. Operating at 5V with a 16 MHz clock, it features 14 digital I/O pins, 8 analog inputs, and 6 PWM outputs, making it versatile for tasks like robotics, IoT, and automation as shown in figure 2.2.

It supports USB programming via a mini or micro USB port, offering easy code uploads and debugging. Its small size, low power consumption, and compatibility with various sensors and modules make it a popular choice for space-constrained and innovative projects in prototyping and embedded systems.

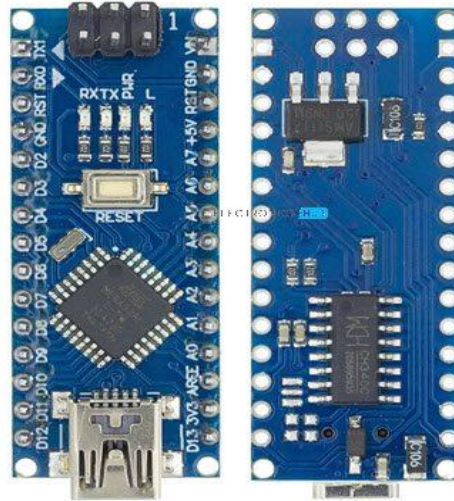


Fig. 2.2 Arduino Nano

The diagram highlights key components and the pin layout of the Arduino Nano, including power pins, I/O pins, and the USB interface, aiding in project design and implementation.

2.4.1 TECHNICAL FEATURES

Table 2.1 shows the technical features of Arduino Nano Board.

Table 2.1 Technical Features of Arduino Nano Board

Microcontroller	ATmega328P
Operating Voltage	5V
Input voltage	7-12V
Digital I/O Pins	14 (6 support PWM output)
Analog Input Pins	8
DC Current per I/O Pin	40 mA
Flash Memory	32 KB (2 KB used by boot loader)
SRAM	2 KB
EEPROM	1 KB
Clock Speed	16 MHz
USB Interface	Mini or Micro USB
Dimensions	45mm x 18mm

2.4.2 POWER

The Arduino Nano is designed to support multiple power options, making it versatile for various applications. Here are the key details about its power capabilities:

- I. **Operating Voltage:** The board operates at 5V, which is internally regulated.
- II. **Input Voltage (recommended):** The recommended range for the input voltage is 7-12V.
- III. **Input Voltage (limit):** The board can handle input voltages between **6-20V**, but prolonged use at extremes may cause damage.
- IV. **Power Sources:**
 - a. **USB Power:** The Nano can be powered directly via the mini or micro USB connection, ideal for programming and small-scale projects.
 - b. **VIN Pin:** The VIN pin allows the board to be powered by an external power source, such as a battery or adapter (7-12V recommended).
- V. **3.3V and 5V Outputs:**
 - a. **5V Pin:** Provides regulated 5V output for powering external components.
 - b. **3.3V Pin:** Offers a regulated 3.3V output for low-voltage devices.
- VI. **DC Current:**
 - a. The Nano can supply a maximum of **40 mA per I/O pin**, making it suitable for driving small LEDs, sensors, or modules.
- VII. **Low Power Consumption:** The board consumes approximately **19 mA** during standard operation, making it ideal for energy-efficient projects.

The flexibility of power options and low energy consumption makes the Arduino Nano a reliable choice for embedded systems, portable devices, and IoT applications.

2.4.3 MEMORY

The **Arduino Nano** is based on the **ATmega328P** microcontroller and has three types of memory: **Flash**, **SRAM**, and **EEPROM**.

- **Flash Memory** (32 KB) stores the program code, with 2 KB reserved for the boot loader.
- **SRAM** (2 KB) is used for temporary data storage during program execution and is cleared when the board powers off.

- **EEPROM** (1 KB) is non-volatile memory used for storing data that must persist across resets or power cycles.

In short, the Nano has 32 KB for program storage, 2 KB for temporary data, and 1 KB for persistent data storage.

2.4.4 INPUT AND OUTPUT

The **Arduino Nano** based on the **ATmega328P** microcontroller features a variety of input and output (I/O) capabilities for interfacing with sensors, actuators, and other devices.

Digital I/O Pins

The Arduino Nano has **14 digital I/O pins** (pins 0 to 13), which can be used for both input and output. These pins can be configured as either input or output, depending on the needs of the application:

- **Input Mode:** These pins can read high or low digital signals (0 or 5V).
- **Output Mode:** They can output digital signals (0 or 5V), controlling LEDs, relays, or other devices.

Additionally, **pins 0 and 1** (used for serial communication) can also serve as digital I/O.

Analog Input Pins

The Arduino Nano provides **8 analog input pins** (pins A0 to A7), which allow it to read analog signals ranging from 0 to 5V. The **10-bit ADC** (Analog-to-Digital Converter) converts these signals into values between 0 and 1023, allowing the board to interface with sensors like temperature sensors, potentiometers, or light sensors.

PWM Output Pins

The Arduino Nano has **6 pins** (pins 3, 5, 6, 9, 10, and 11) that can provide **PWM** (Pulse Width Modulation) output. These pins allow for simulating an analog output by rapidly switching the signal on and off, which can be useful for controlling the brightness of LEDs or the speed of motors.

Serial Communication

The board has built-in support for **serial communication** through pins 0 (RX) and 1 (TX). These are used for transmitting and receiving data to and from other devices, such as sensors or computers, for debugging or communication with external systems.

Power Pins

- **5V and 3.3V Pins:** The Nano provides 5V and 3.3V pins to power external components.
- **GND Pins:** Multiple GND (ground) pins are available for circuit connections.
- **RAW Pin:** If using an external power source, the RAW pin can supply voltage between 6-12V, which is regulated to 5V on the board.

2.4.5 PROGRAMMING

The **Arduino Nano** can be programmed using the **Arduino IDE** (downloadable from the official website). The **ATmega328P** on the Arduino Nano comes pre-programmed with a boot loader that enables you to upload new code without the need for an external hardware programmer. It communicates using the original **STK500 protocol** for easy uploading of sketches through the USB connection. For more details, refer to the official Arduino documentation and tutorials.

2.5 SIM 800L

The **SIM800L** is a compact and cost-effective GSM/GPRS (Global System for Mobile Communications/General Packet Radio Service) module that allows microcontrollers to communicate over cellular networks. It is widely used in IoT (Internet of Things) projects and other applications requiring wireless communication.

2.5.1 KEY FEATURES

- I. **GSM/GPRS Support:** Enables sending and receiving SMS, making voice calls, and accessing the internet via GPRS (2G).
- II. **Quad-Band Support:** Operates on GSM 850/900/1800/1900 MHz, making it compatible with global cellular networks.

- III. **Compact Design:** Small in size (25mm × 23mm), making it ideal for projects with limited space.
- IV. **Low Power Consumption:** Operates efficiently, consuming low power in idle mode and slightly more during active communication.
- V. **Voltage Range:** Operates at 3.7V to 4.2V, typically powered by a Li-Po battery or a regulated power supply.
- VI. **Communication Interface:** Communicates with microcontrollers like Arduino, ESP8266, or Raspberry Pi via UART (serial communication).
- VII. **Antenna Options:** Comes with a connector for an external antenna, enhancing signal strength in areas with weak coverage.

2.5.2 APPLICATIONS

- Sending/receiving SMS and making calls.
- Remote monitoring and control.
- IoT projects, such as home automation or smart meters.
- Vehicle tracking and GPS integration (when combined with a GPS module).

2.5.3 WORKING

The SIM800L is controlled via **AT commands**, which are instructions sent to the module over a serial interface to perform specific tasks like sending SMS, making calls, or connecting to the internet. For example:

- To send an SMS: Use the command `AT+CMGS="phone_number"`.
- To make a call: Use `ATDphone_number;`.

The **SIM800L** is a versatile module ideal for adding cellular connectivity to various embedded systems.

2.5.4 PIN DESCRIPTION

VCC - Power input (3.7V–4.2V). Use a stable power source capable of 2A.

GND - Ground connection.

RXD - Receive pin for serial data (connect to Arduino TX).

- Voltage: 3.3V logic.

TXD - Transmit pin for serial data (connect to Arduino RX).

-Voltage: 3.3V logic output.

RST - Reset pin (active low). Pull low to reset the module.

NET (LED) - Network status indicator:

- Blinks every second: Searching for a network.
- Blinks every 3 seconds: Connected to a network.

Antenna - Connects to an external antenna for better signal reception.



Fig. 2.3 SIM 800L

2.5.5 INTERFACING SIM 800L

Interfacing the **SIM800L GSM module** with an **Arduino Nano** enables features like sending SMS, making calls, and accessing GPRS. The **SIM800L** is powered by a 3.7–4.2V supply, with its **TXD** connected to the Arduino's **RX** (D2) and **RXD** to **TX** (D3) via Software Serial.

The Arduino communicates with the SIM800L using **AT commands**, such as **AT+CMGF=1** for SMS mode and **AT+CMGS** to send messages. A stable power supply (2A) is crucial for reliable operation. This setup is ideal for IoT applications like notifications, remote monitoring, and tracking.

2.6 LIQUID CRYSTAL DISPLAY (LCD)

Liquid Crystal Display (LCD) screen is an electronic display module which finds a wide range of applications. 16x2 LCD display is a very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being, LCDs are economical, easily programmable, have no limitations of displaying special and even custom characters (unlike in seven segments), animations and lot of other useful features. A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, command and data. The command register stores the command instructions given to the LCD. A command is an instruction given to the LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD. Pin diagram of LCD is shown in figure 2.4 and details of its pin configuration are given Table 2.2.

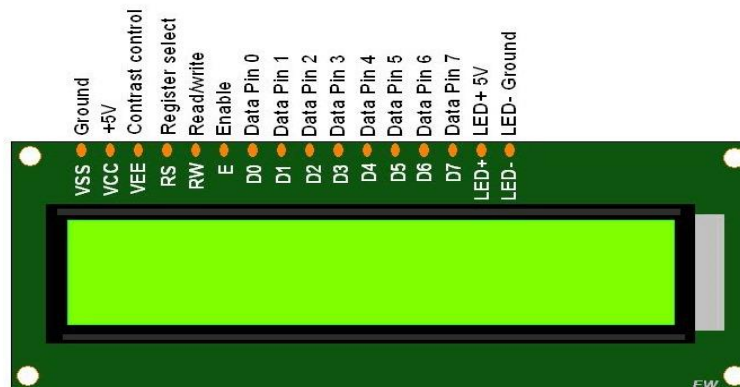


Fig.2.4 LCD Pin Diagram

Table 2.2 Pin Configuration of LCD

PIN NO.	SYMBOL	I/O	DESCRIPTION
1	VSS	--	Ground
2	VDD	--	+5V power supply
3	VE	--	Power supply to control contrast
4	RS	I	RS=0 to select command register RS=1 to select data register
5	R/W	I	R/W=0 for write R/W=1 for read
6	EN	I/O	Enable
7	DB0	I/O	The 8-bit data bus
8	DB1	I/O	The 8-bit data bus
9	DB2	I/O	The 8-bit data bus
10	DB3	I/O	The 8-bit data bus
11	DB4	I/O	The 8-bit data bus
12	DB5	I/O	The 8-bit data bus
13	DB6	I/O	The 8-bit data bus
14	DB7	I/O	The 8-bit data bus
15	+LED	--	LED Back light anode
16	-LED	--	LED Back light cathode

2.7 BUZZER

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers and confirmation of user input. Electronic buzzer as shown in the figure 2.5.



Fig. 2.5 Electronic Buzzer

2.8 POTENTIOMETER

A potentiometer, often called a "pot," is a three-terminal resistor with an adjustable sliding or rotating contact. It acts as a variable resistor or voltage divider. Potentiometers are commonly used to adjust levels such as volume in audio devices, control brightness, or tune circuit parameters as shown in the following figure 2.6.

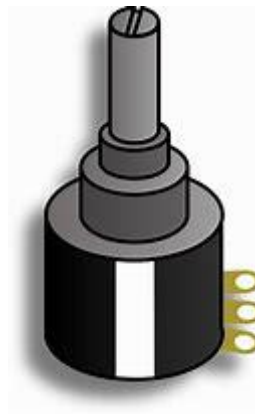


Fig. 2.6 Potentiometer

In our project, the potentiometer likely serves one or more of the following purposes:

- **Adjusting Input Voltage:** It can control the input voltage to a circuit, such as a sensor or the SIM800L module, ensuring precise adjustments.
- **Signal Control:** If the project involves analog signals, the potentiometer can be used to fine-tune the signal levels for better performance.
- **User Control Interface:** The potentiometer might act as a user input device, such as adjusting brightness, sound, or thresholds for sensor readings.
- **Testing and Calibration:** The potentiometer could be included to allow easy calibration of parameters like voltage or signal strength during development or testing.

By adjusting the potentiometer, we can dynamically control aspects of the circuit, making it a versatile component in many electronic projects.

2.9 BUCK CONVERTOR

A **buck potentiometer** typically refers to a **potentiometer** used in conjunction with a **buck converter**. A **buck converter** is a type of DC-DC power converter that steps down a higher

input voltage to a lower output voltage. In this context, the potentiometer is used as a control interface to adjust the output voltage or current of the buck converter as shown in figure 2.7.



Fig. 2.7 Buck Convertor

2.9.1 WORKING

- The potentiometer acts as a variable resistor or a voltage divider.
- By turning the knob or sliding the wiper, the resistance changes, which alters the feedback voltage provided to the buck converter's controller.
- The controller adjusts the duty cycle of the converter's switching mechanism to regulate the output voltage accordingly.

2.10 EM18 MODULE READER

The EM18 Module Reader is a commonly used RFID (Radio Frequency Identification) module designed to read RFID tags and cards. It operates at a frequency of 125 kHz and is widely used in applications such as access control, attendance systems, and other RFID-based systems.

2.10.1 PIN DESCRIPTION

- **VCC:** Power supply (5V).
- **GND:** Ground.
- **TX:** Serial output for sending the tag/card ID to a microcontroller.
- **RST:** Reset pin to restart the module.
- **BEEP:** Optional pin to control a buzzer for feedback.

- **LED:** Can be connected to an LED to indicate successful tag/card detection.

2.10.2 FEATURES

- I. **Operating Frequency:** 125 kHz.
- II. **Interface:**
 - a. **Serial (UART):** Outputs the card ID as ASCII via TX pin.
 - b. **Wiegand:** Outputs data in Wiegand format for specific applications.
- III. **Range:** Typically 5–10 cm (depending on the RFID tag/card and antenna).
- IV. **Power Supply:** Operates at 5V.
- V. **Output Data:** Provides the unique ID of the RFID tag/card when scanned.

2.10.3 HOW RFID TAGS ARE USED WITH EM 18 MODULE READER

The EM18 RFID module is designed to read 125 kHz passive RFID tags, which are powered by the electromagnetic field generated by the module. Here's how RFID tags interact with the EM18 reader we can see in the following figure 2.8.

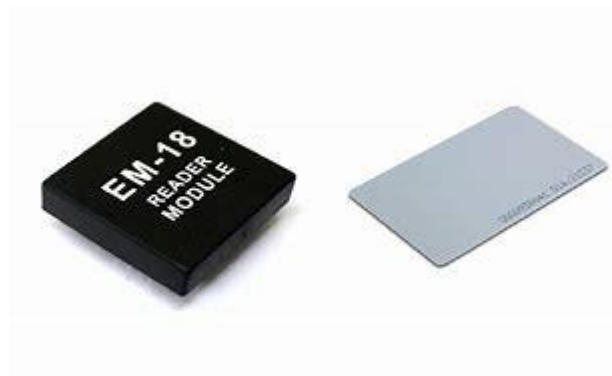


Fig. 2.8 EM 18 Module Reader with RFID Tags

Process of RFID Tag Usage in EM18:

- I. **Powering the RFID Reader:** The EM18 module is powered by a 5V supply, which activates its internal antenna to emit a 125 kHz electromagnetic field.
- II. **Tag Detection:** When a passive RFID tag (e.g., a card or fob) is brought close to the EM18 module, the tag's antenna captures the electromagnetic energy, powering its internal chip.
- III. **Tag Data Transmission:** The powered RFID tag responds by transmitting its unique **ID number** back to the EM18 reader via radio waves.

- IV. **Data Processing:** The EM18 module receives the tag's data, processes it, and converts it into a readable format (usually ASCII or hexadecimal).
- V. **Output via UART:** The processed data (the unique ID of the RFID tag) is sent to a connected microcontroller (e.g., Arduino Nano) or system via the **TX pin** of the EM18 module using UART communication.
- VI. **System Action:** The microcontroller uses the tag's unique ID to perform predefined actions, such as granting access, logging attendance, or triggering an alert.

2.11 LEDs (LIGHT EMITTING DIODE)

An LED (Light Emitting Diode) is a semiconductor device that emits light when an electric current flows through it. LEDs are energy-efficient, long-lasting, and widely used in various applications, from indicators to illumination. The internal structure and different shapes of a LED are shown in Fig. 2.9 (a) and (b) respectively.

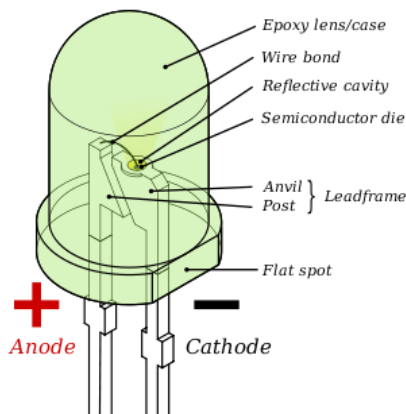


Fig. 2.9(a) Inside a LED

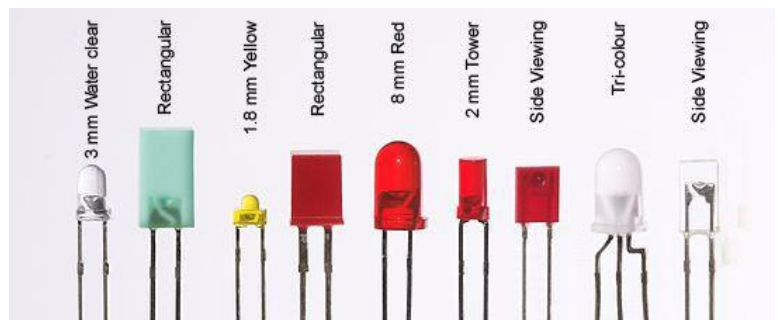


Fig. 2.9(b) Different shapes of LED

2.11.1 WORKING OF LED

By using electroluminescence to emit light, a Light Emitting Diode (LED) operates. The n-type semiconductor's electrons recombine with the p-type region's holes when the LED is forward-biased, producing energy in the form of photons (light). Depending on the semiconductor material's energy band gap, the light's hue changes. Because they effectively transform electrical energy into light, materials like gallium arsenide (GaAs) and gallium phosphide (GaP) are employed and utilized. LEDs are incredibly energy-efficient and produce

light nearly quickly when powered, in contrast to conventional diodes. The various advantages of LED are listed as follows:

- Smaller in size
- Light in weight
- Mechanically rugged
- Lower operating temperature
- Small energy consumption
- On time is very small

CHAPTER THREE

EXECUTION OF THE PROJECT HARDWARE

3.1 IMPLEMENTATION OF THE HARDWARE

The hardware development process for the entire suggested system is described in detail in this chapter. Building a working system that combines the EM18 RFID module and Arduino Nano to read RFID tags and work with the data to accomplish the intended result—for example, allowing access, recording data, or managing linked devices—is the primary goal.

3.2 CIRCUIT DIAGRAM OF THE HARDWARE

Fig. 3.1 displays the circuit diagram for the RFID-based diagram. Together along with the EM18 RFID module, which was covered in the previous chapter, the Arduino Nano serves as the system's central component in this hardware model. Using the information it receives from the module, the circuit is made to read data from RFID tags and carry out operations like data logging or access control.

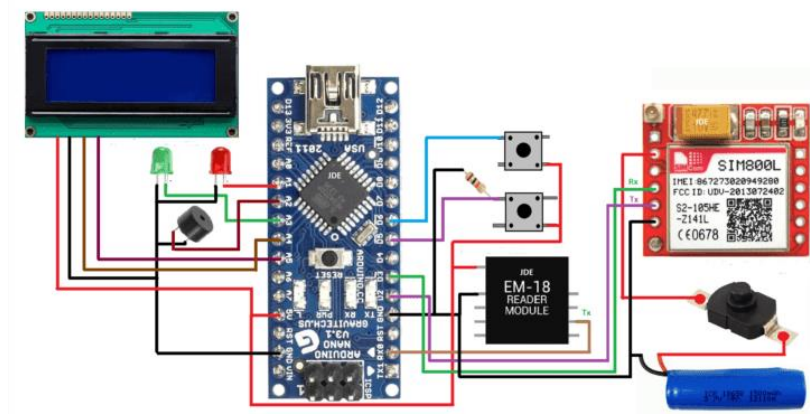


Fig. 3.1 Circuit Diagram for Smart trolley with automatic billing system

3.3 WORKING

The project is designed to use the EM18 RFID module and Arduino Nano to read RFID tags and perform specific actions based on the data received. Below is the step-by-step working of the system:

- Power supply: The system is powered by a 5V DC source that is controlled. For the Arduino Nano and EM18 RFID module to function, a voltage regulator makes sure they both receive a steady voltage.
- Initialization: The Arduino Nano establishes communication with the EM18 RFID module and initializes the system when it is turned on. It also readies the linked parts (such relays, displays, or LEDs) for use.
- RFID Tag Detection: The EM18 RFID module creates a high-frequency electromagnetic field for RFID tag detection. The RFID tag activates and sends its unique ID to the EM18 module when it enters this field's range.
- Data Transmission: Through a serial UART connection, the EM18 module takes the tag's unique ID and transmits it to the Arduino Nano.
- Data processing: The Arduino Nano receives the tag data and uses its memory to compare it with previously recorded values. Predetermined actions are taken by the system based on whether the tag's ID matches or does not match.
- Output Activities:
 - The device does the required action, such turning on an LED, allowing access, or displaying a message on an LCD screen, if the tag ID matches a value that has been saved.
 - A red LED or buzzer may be activated by the system as an error indication if the tag ID is not recognized.
- Real-Time Feedback: Feedback mechanisms, such as LEDs or an LCD display, provide real-time status updates. For example:
 - A green LED lights up for authorized tags.
 - A red LED or buzzer is activated for unauthorized tags.
- Continuous Operation: The system runs in a never-ending loop, ready to recognize and process new RFID tags as they approach.

This chapter described the hardware implementation of the project, including the integration of the Arduino Nano, EM18 RFID module, and auxiliary parts. The system's functionality is based on the hardware configuration. We shall talk about the project software's execution in the upcoming chapter, with an emphasis on system operation and programming.

CHAPTER FOUR

EXECUTION OF THE PROJECT SOFTWARE

4.1 DOWNLOADING ARDUINO IDE (ARDUINO PROGRAMMING SOFTWARE)

We can upload new programs to the Arduino Nano using the **Arduino IDE**, an open-source platform that does not require a license. The Arduino IDE is compatible with **Windows**, **Mac OS X**, and **Linux** operating systems. The latest version of the Arduino IDE can be downloaded from the following link:

➤ <http://arduino.cc/en/Main/Software>

To use the Arduino software, first download and extract it to a preferred location. To create a new project, double-click the **arduino.exe** icon, which can be found on the desktop or in the start menu. This opens the Arduino IDE. Next, select the appropriate board for the project by navigating to **Tools > Board > Arduino Nano**, as shown in Fig. 4.1. This ensures the IDE is correctly configured for programming the Arduino Nano in the system.

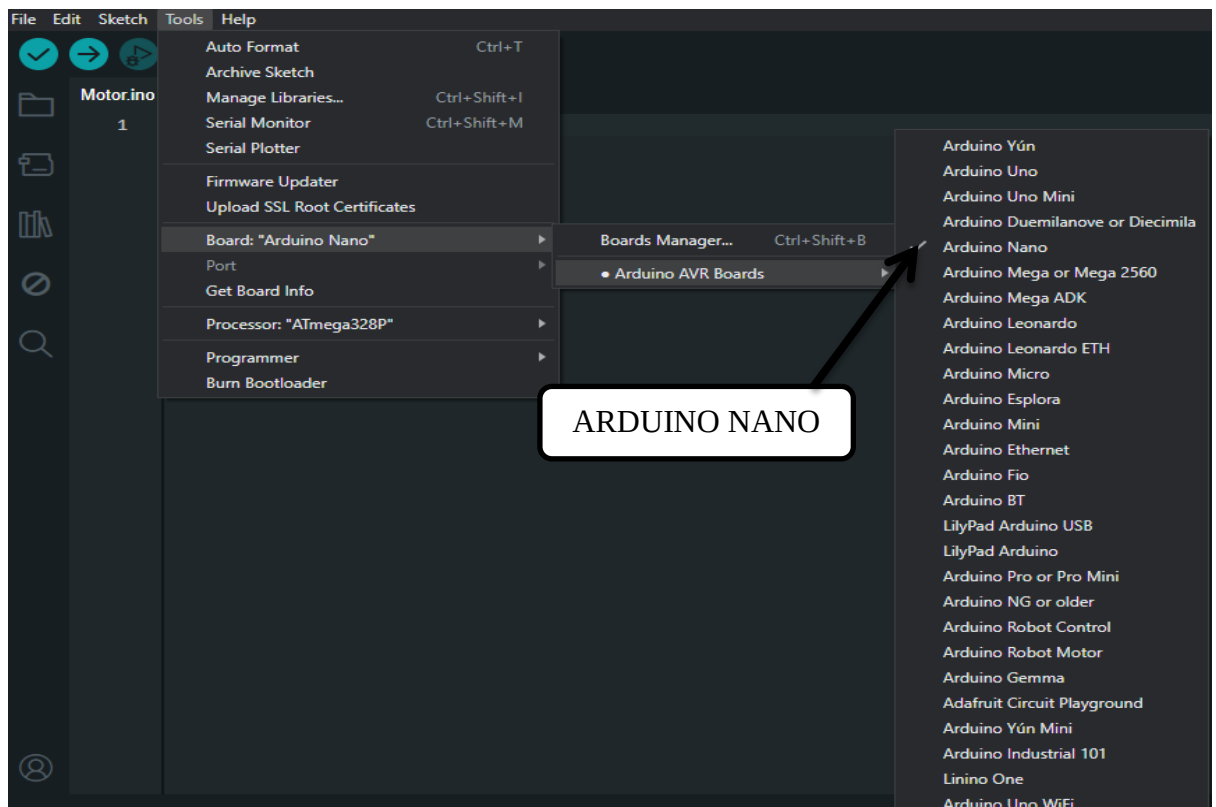


Fig. 4.1 Selection of Arduino Nano

We may also select the COM port for the Arduino Nano as shown in **Fig. 4.2**. To do this, navigate to **Tools > Port > COM3** (or any available COM port, depending on the PC). This step ensures proper communication between the Arduino Nano and the computer.

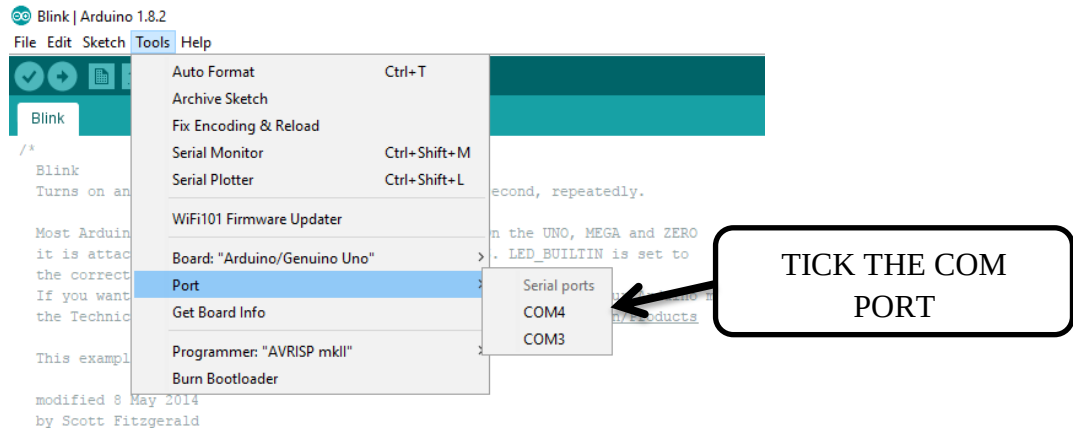


Fig. 4.2 Selection of COM Port

4.2 UPLOADING AN ARDUINO SKETCH (PROGRAM)

After Connecting Arduino pro mini to computer via burner, first of all we will upload the most basic example, which is blinking an on-board LED using digital pin 13. Open Arduino IDE. Go to **File > Examples > 01.Basics > Blink** as shown in Fig. 4.3.

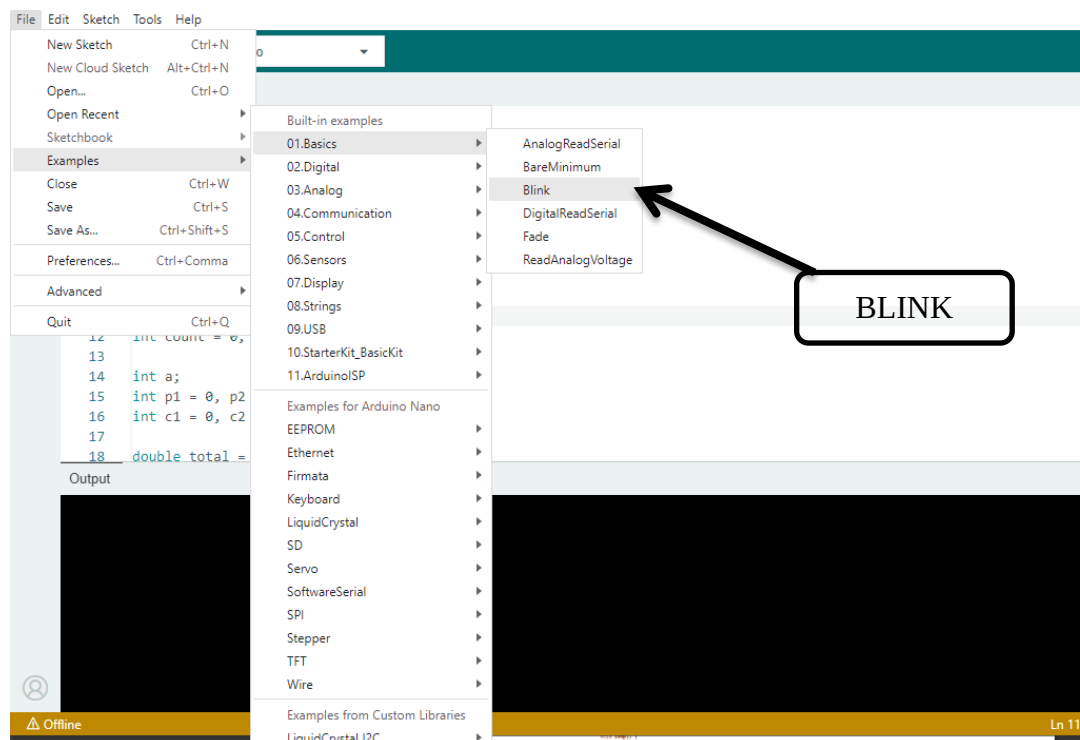


Fig. 4.3 Selection of LED Blink Program on Digital Pin 13

By default, Arduino IDE is available pre-configured for the Arduino pro mini. After compiling the sketch (program) click the “Upload” button and wait a few seconds until a “Done uploading” message appears as shown in Fig. 4.4.

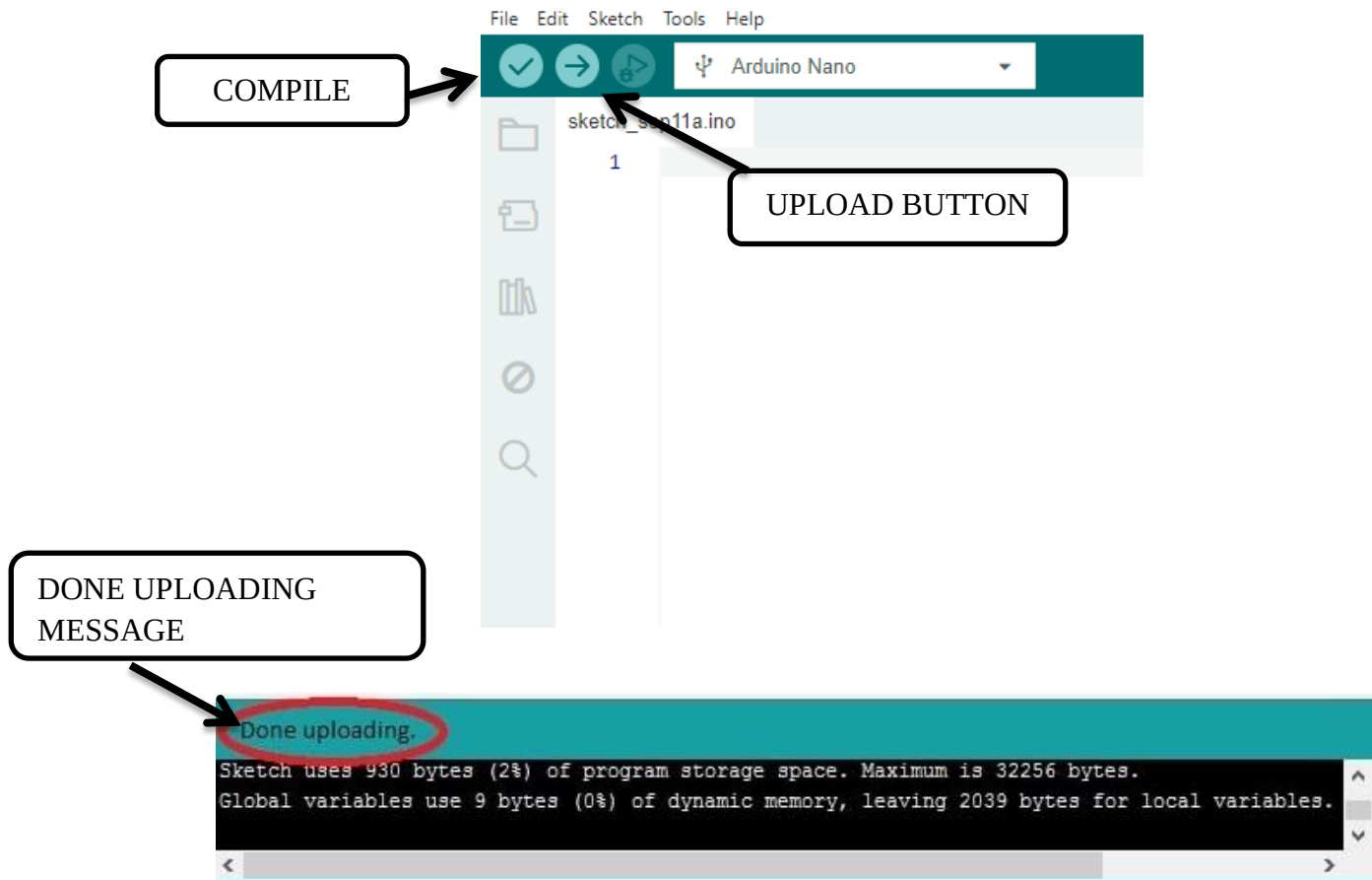


Fig. 4.3 Compilation and Uploading the Program on to the Arduino Nano

This code simply blinks on-board LED on Arduino pro mini, highlighted with red color as shown in Fig. 4.4. If we look closely we should see the LED staying on for one second and off for another second repeatedly.

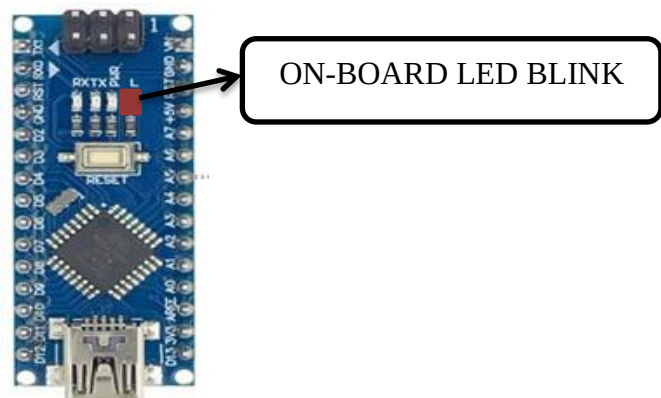


Fig. 4.4 Blinking of On-Board LED for the Arduino Nano

The developed hardware program is burnt in a similar way as explained earlier for “blink” program on Arduino Pro Mini through Arduino IDE. This programming language is very DONE UPLOADING MESSAGE COMPILE BUTTON UPLOAD BUTTON ON-BOARD LED BLINK FOR EVERY 1 SEC. 28 much similar to C language however Arduino provides specific libraries, used only in this environment.

4.3 Arduino Overview

Arduino is a popular programmable board used to create projects. It consists of a simple hardware platform as well as a free source code editor which has a “one click compiles or upload” feature. Hence it is designed in way that one can use it without necessarily being an expert programmer (Kushner 1987). Arduino offers an open-source electronic prototyping platform that is easy to use and flexible for both the software and hardware. Arduino is able to sense the environment through receiving input from several sensors. It is also able to control its surrounding through controlling motors, lights and other actuators. The Arduino programming language that is based on the wiring and the Arduino development environment that is based on the processing are used to program the microcontroller found on the board (Banzi, 2005). Due to its open-source environment, one is able to easily write and upload codes to the I/O board. It is also worth to note that Arduino can be run on Linux, Mac OSX and Windows as its environment is written in Java.

4.4 Arduino Development

While discussing the development of Arduino, it is worth introducing a brief history of microcontrollers. A revolutionary leap in the computing industry was seen in the 1960s following the development of solid state computers (including the IBM 1401), that used transistors to process its operations and a magnetic core memory for its storage (instead of vacuum tubes), and these enabled an increase in the compactness of the computer hardware. In addition, Jack Kilby’s invention of integrated circuits in 1959 enabled circuits and transistors to be fused into tiny chips of semiconducting materials (like silicon) as well as further miniaturization of the computer component. The other crucial development made in the same decade was the high-level computer programming languages, written in symbolic languages such as plain English, and this made computer codes somehow easy to learn and read than the earlier machine languages that consisted of letters and numbers only. This development enabled individuals with few years of expertise to carry out the basic operations

on a computer. FORTRAN (for the scientific calculators) and COBOL (for business application) were the two main languages that were introduced in that period. The microprocessor was one of the greatest innovations in the history of the modern computer in the 1970's. Initially, the microprocessor miniaturized all the hardware components of CPU to fit into one, tiny, integrated circuit, popularly known as the microchip. The microchip became the major driving component of the microcontrollers including Arduino which is made up of a microchip, input/output hardware and memory storage hardware for sensors. The microprocessor, due to the small form factor, was incorporated into a surfeit of electronic devices ranging from personal computers to calculators and are still used up to date. More programming languages were also developed in the 1970s and 80s including C, C++ and Java for applications in science and business.

4.5 Why Arduino Boards?

Arduino board has been used for making different engineering projects and different applications. The Arduino software is very simple to use for beginners, yet flexible adequate for advanced users. It runs windows, Linux and Mac. Teachers and students in the schools utilize it to design low cost scientific instruments to verify the principles of physics and chemistry. There are numerous other microcontroller platforms obtainable for physical computing. The Netmedia's BX-24, Parallax Basic Stamp, MIT's Handyboard, Phidget and many others present related functionality. Arduino also makes simpler the working process of microcontroller, but it gives some advantages over other systems for teachers, students and beginners.

=> Inexpensive

=> Cross-platform

=> Simple, clear programming environment

=> Open source and extensible software

=> Open source and extensible hardware

In order to demonstrate how simple Arduino boards are to utilize for RFID-based projects, this chapter covered their features, uploading sketches, and setting up the IDE. Using the EM18 module, the RFID-based system's analysis and outcomes will be covered in the next chapter.

CHAPTER FIVE

RESULTS, DISCUSSION AND ANALYSIS

5.1 INTRODUCTION

This hardware offers a practical and efficient way to use the EM18 module in an RFID-based system. The use of affordable and easily accessible components is ensured by the circuit's design, which prioritizes simplicity. Its simple design makes it affordable and simple to put together.

We conducted real-time system performance tests. The Arduino Nano reads and processes the RFID tag's unique ID when it comes into contact with the EM18 module. A green LED turns on and the status is shown on the LCD if the tag ID matches the data that has been stored. The LCD shows the unauthorized access, and a red LED or buzzer warns the user if the tag is not recognized.

The RFID reader and Arduino Nano actively monitor tags as the system runs smoothly in continuous mode. The LCD shows the permission status of the identified tags and their unique IDs for real-time feedback.

A successful hardware model of the RFID-based system has been created, as Fig. 5.1 illustrates. Applications like inventory management, attendance monitoring, and access control show how useful this technology is.



Fig. 5.1 (a) Pictures of Hardware Model

When I scan the product RFID Tag the LCD will display the product name and price of the product.in 3 and 4 lines Display the total price in the Fig. 5.1(b).



Fig 5.1(b) Picture of Hardware Models

If I try to remove an item then just press and hold the green push button and scan the product RFID Tag the LCD will display the removed item message and decrease the total cost as we can see in the Fig 5.1(c).



Fig. 5.1(c) Pictures of Hardware Models

Now if I press the yellow push button then GSM Send the Text message or Proper Bill to the mobile number if you enter your UPI Number then you can pay it, we can see in the Fig 5.1(d).



Fig. 5.1(d) Pictures of Hardware Models

5.2 COST CALCULATION OF THE HARDWARE MODEL

The cost of the hardware model implemented has been given in Table 5.1.

S.NO.	COMPONENT(S)	QUANTITY	RATE	AMOUNT IN RS.
1	Arduino Nano	1	250	250
2	SIM 800L	1	280	280
3	EM-18RFID	1	280	280
4	20 X 4 LCD	1	380	380
5	EM-18 Tag	4	25	100
6	Red LED	1	1	1
7	Green LED	1	1	1
8	Buzzer	1	15	15
9	I2C Module	1	80	80
10	Button	2	10	20
11	3.7 V Battery	1	50	50
12	Male to Male Wires	10	2	20
13	Male to Female wires	10	1	10
14	Resistance(1 Ω ,5 Ω)	--	5	5
			Total	1582

CHAPTER SIX

CONCLUSIONS AND FUTURE SCOPE

6.1 CONCLUSION

This project effectively uses the Arduino Nano and EM18 module to create an RFID-based system. Applications such as access control and records of attendance might benefit from the design's simplicity, affordability, and dependability. It effectively reads and processes data from RFID tags to carry out preset tasks, demonstrating the usefulness of RFID technology. In addition to accomplishing its goals, the initiative lays the groundwork for further improvements.

6.2 FUTURE SCOPE

The proposed hardware, along with some modification can be used for the purpose such as;

- **Integration with Passive Infrared Sensor:** The system can be expanded by integrating a passive infrared sensor with a Fresnel lens to monitor a 360° area. This would enable the fan to turn on or off automatically based on the motion of people, enhancing energy efficiency and convenience.
- **Manual and Automatic Control with Changeover Switch:** A changeover switch could be added to the system, allowing the fan to be controlled both manually and automatically. This provides flexibility for users to choose their preferred mode of operation.
- **Application in Heater Circuits:** With some modifications, the temperature-controlled fan can be used in heater circuits to maintain a constant temperature. This could improve safety and performance in devices that require temperature regulation.
- **Alarm System Integration for Overheating:** The system can be further developed by incorporating an alarm circuit. This would be especially useful in large equipment where overheating and the risk of explosions are serious concerns, particularly in industries where machinery needs constant monitoring for safety.

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